4. For every CO_2 molecule entering the C_3 cycle, the number of ATP & NADPH required
   a. 2ATP + 2NADPH
   b. 2ATP + 3NADPH
   c. 3ATP + 2NADPH
   d. 3ATP + 3NADPH
5. Identify true statement regarding light reaction of photosynthesis.
   a. Splitting of water molecule is associate with PS I.
   b. PS I and PS II involved in the formation of NDPH+H^{+}.
   c. The reaction center of PS I is Chlorophyll a with absorption peak at 680 nm.
   d. The reaction center of PS II is Chlorophyll a with absorption peak at 700 nm.
6. Two groups (A & B) of bean plants of similar size and same leaf area were placed in identical conditions. Group A was exposed to light of wavelength 400-450nm & Group B to light of wavelength of 500-550nm. Compare the photosynthetic rate of the 2 groups giving reasons.
7. A tree is believed to be releasing oxygen during night time. Do you believe the truthfulness of this statement? Justify your answer by giving reasons.
8. Grasses have an adaptive mechanism to compensate photorespiratory losses- Name and describe the mechanism.
9. In Botany class, teacher explains, Synthesis of one glucose requires 30 ATPs in C_4 plants and only 18 ATPs in C_3plants. The same teacher explains C_4 plants are more advantageous than C_3 plants. Can you identify the reason for this contradiction?
10. When there is plenty of light and higher concentration of O_2, what kind of pathway does the plant undergo?Analyse the reasons.



ICT Corner

Photosynthesis

Let's play **photosynthesis**

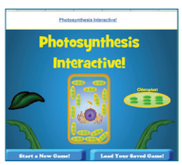


Steps

- Scan the QR code
- Start a new game and tap
- Click light dependent reaction and follow the steps
- After completion – move back and Click Calvin cycle reaction and follow the steps

Activity

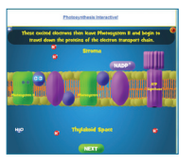
- Observe the cycle and record it
- Check your grasp by click the Quiz tap
- Conclude your observations.



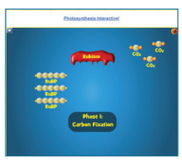
Step 1



Step 2



Step 3



Step 4

LE2DZ

Web URL:
https://biomanbio.com/HTML5GamesandLabs/PhotoRespgames/photointeractivehtml5page.html

* Pictures are indicative only

11th Botany_EM_Chapter 13.indd 290 06/01/2023 20:02:56